KENYA MEDICAL TRAINING COLLEGE
FACULTY OF REHABILITATIVE HEALTH SCIENCE
DEPARTMENT OF MEDICAL ENGINEERING

FINAL QUALIFYING EXAMINATION
FOR
CERTIFICATE IN MEDICAL ENGINEERING TECHNOLOGY

PAPER: ELECTRONICS

DATE: 1st February, 2023 **TIME: 3 HOURS (9:00 A.M – 12 NOON)**

INSTRUCTIONS

1. This paper consists of:
   - Section 1 (40 Multiple Choice Questions)
   - Section 2 (8 Short Answer Questions)
   - Section 3 (1 Long Answer Question)
2. Attempt **ALL** Questions
3. All multiple choice questions **MUST** be answered in **CAPITAL LETTERS**
4. Write the **EXAMINATION NUMBER** given on all the answer sheets provided and on the question paper.
5. Ensure that all examination answer scripts are handed in at the end of the examination
6. Ensure you sign the examination register provided

EXAMINATION NUMBER .....

ELECTRONICS

SECTION 1: MULTIPLE CHOICE QUESTIONS (40 MARKS)

1. What is the value of a resistor whose colour code is orange, orange, black, yellow and silver?
   A. 330 KΩ
   B. 3.3 MΩ
   C. 3.3 KΩ
   D. 33 KΩ
2. Which one of the following is not a factor considered when selecting a capacitor?
   A. Tolerance
   B. Stability
   C. Leakage current
   D. Capacitive reactance
3. A trivalent impurity is added to silicon to create
   A. Germanium
   B. P-type semiconductor
   C. N-type semiconductor
   D. A depletion region
4. In the forward region of its characteristics, a diode appears as
   A. An off switch
   B. A high resistance
   C. A capacitor
   D. An on switch
5. For a junction FET in the pinch – off region when the drain voltage is increased, the drain current
   A. Becomes zero
   B. Abruptly decreases
   C. Abruptly increases
   D. Remains constant
6. Which MOSFET allows the drain current to flow even with zero gate to source voltage?
   A. Depletion MOSFET
   B. Enhancement MOSFET
   C. Depletion – enhancement MOSFET
   D. All of the above
7. When the BJT is switched on, it is said to be
   A. Shorted
   B. Open
   C. Saturated
   D. Cut-off

8. In a BJT, the largest current flows
   A. In the emitter
   B. In the collector
   C. In the base
   D. Through the collector base junction
9. In properly biased BJT, an increase in base current causes increase in
   A. I_C only
   B. I_E only
   C. Both I_C and I_E
   D. Leakage current
10. The emitter of a transistor is generally heavily doped because it
   A. Has to dissipate maximum power
   B. Has to supply charge current
   C. It is the first region of the transistor
   D. Must possess low resistance
11. The dc load line of a transistor circuit
   A. Has a negative slope
   B. It is a curved line
   C. Gives graphic relationship between I_B & I_C
   D. Does not contain a point
12. After firing an SCR, the gate current is removed. The current in the SCR will
   A. Remain the same
   B. Immediately falls to zero
   C. Rise up
   D. Rise a little bit and then fall to zero
13. Triac is a semiconductor power electronic device which contains
   A. Two SCRs connected in reverse parallel
   B. Two SCRs connected in parallel
   C. Two SCRs connected in series
   D. Two BJTs connected in series
14. _____ is a self-generation type transducer that can produce an electrical voltage or current output without utilizing any external power source.
   A. A passive transducer
   B. An active transducer
   C. A primary transducer
   D. A secondary transducer
15. Chose the correct statement
   A. MOSFET is a unipolar voltage controlled two terminal device
   B. MOSFET is a bipolar current controlled three terminal device
   C. MOSFET is unipolar voltage controlled, three terminal device
   D. MOSFET is a bipolar, current controlled two terminal device

16. Amplifier stages is dc amplifier are?
    A. Directly coupled
    B. Transformer coupled
    C. RC coupled
    D. Impedance coupled
17. Why is the circuit of DC amplifier cheaper?
    A. More resolutions used
    B. More components accommodated
    C. Complex
    D. No expensive coupling devices
18. _____ multi-vibrator is a square wave oscillator.
    A. Monostable
    B. Astable
    C. Bistable
    D. None of the above
19. A circuit which consist of quasi stable state is called?
    A. Bistable
    B. Monostable
    C. Tristable
    D. Astable
20. Multi-vibrators belong to a family oscillator commonly called?
    A. Relaxation oscillators
    B. Dynamic oscillators
    C. Sinusoidal oscillator
    D. Harmonic oscillator
21. Multi-vibrator which can produce very short pulse or a much longer rectangular shaped wave form is called
    A. Astable
    B. Bistable
    C. Monostable
    D. Tristable
22. _____ multi-vibrator can be used as an electronic delay line.
    A. Monostable
    B. Astable
    C. Bistable
    D. All of the above
23. An oscillator converts _____
    A. AC power to DC
    B. DC power to AC
    C. Mechanical power to AC
    D. AC power to mechanical power

24. An oscillator produces _____ oscillators.
    A. Damped
    B. Undammed
    C. Modulated
    D. None of the above
25. An oscillator employs _____ feedback.
    A. Positive
    B. Negative
    C. Positive and negative
    D. None of the above
26. Which of the following is correct for digital circuit?
    A. Less susceptible to noise
    B. Uses transistor to create logic gates
    C. Easier to perform error detection
    D. All of the above
27. _____ are basic logic gates.
    A. NOT, OR and NOR
    B. NAND, NOR and AND
    C. AND, OR and EXOR
    D. NOT, AND, and OR
28. _____ are universal gates.
    A. NOT and OR
    B. NAND and NOR
    C. AND and OR
    D. NOT, AND & OR
29. The output of an AND gate with three inputs A, B & C is high when _____
    A. A=1, B=1 and C=0
    B. A = 0, B = 0 and C= 0
    C. A=1, B=1 and C=1
    D. A = 1, B=0 and C=1
30. If a 3-input NOR gate has eight input possibilities, how many of those possibilities will result in a HIGH output?
    A. 1
    B. 2
    C. 7
    D. 8
31. What logic function is produced by adding inverters to the inputs of an OR gate?
    A. NOR
    B. NAND
    C. AND
    D. XNOR

32. The output of a OR gate is high
    A. Only when all the inputs are low
    B. Only when all the inputs are high
    C. Only when at least one input is high
    D. Only when at least one input is low
33. An AND gate can be represented as
    A. Switches connected in series
    B. Switches connected in parallel
    C. Transistors connected in series
    D. Transistors connected in parallel
34. An OR gate can be represented as
    A. Switches connected in series
    B. Switches connected in parallel
    C. Transistors connected in series
    D. Transistors connected in parallel
35. Convert the binary equivalent 10101 to its decimal equivalent
    A. 21
    B. 12
    C. 22
    D. 31
36. An oscillator differs from an amplifier because
    A. Has more gain
    B. Requires no input signal
    C. Requires no dc supply
    D. Always has the same input
37. The V – I characteristics for a triac in the first and third quadrants are essentially identical to those of
    A. Transistor
    B. SCR
    C. VJT
    D. MOSFET
38. An ideal operational amplifier has _____
    A. Infinite output impedance, infinite bandwidth, infinite CMRR
    B. Zero output impedance, zero bandwidth, zero CMRR
    C. Zero output impedance, infinite bandwidth, zero CMRR
    D. Zero input impedance, infinite bandwidth, infinite CMRR
39. In a PN junction with no external voltage, the electric field between the P and N material is called _____
    A. Peak
    B. Barrier
    C. Threshold
    D. Path

40. Recombination is when
    A. An electron falls into a hole
    B. A positive and negative ion is bound together
    C. A valence electron become a conduction electron
    D. A crystal is formed

SECTION 2: SHORT ANSWER QUESTION (40 MARKS)

41. a) Draw correctly biased circuits for:
    i. Common emitter configuration **(2 marks)**
    ii. Common collector configuration **(2 marks)**
    b) What is the other name for common collector configuration? **(1 mark)**

42. Explain the following in respect to semiconductors:
    i. Depletion region **(1 mark)**
    ii. Zener breakdown **(2 marks)**
    iii. Avalanche breakdown **(2 marks)**

43. Explain five differences between SCR and Triac. **(5 marks)**

44. State any five applications of multi-vibrators. **(5 marks)**

45. State five differences between oscillators and amplifiers. **(5 marks)**

46. State five differences between analog and digital electronics. **(5 marks)**

47. Convert:
    i. 42_{10} to binary **(2.5 marks)**
    ii. 1001101_2 to decimal **(2.5 marks)**

48. Draw the symbols of the following logic gates:
    i. OR **(1 mark)**
    ii. AND **(1 mark)**
    iii. NAND **(1 mark)**
    iv. NOR **(1 mark)**
    v. NOT **(1 mark)**

SECTION 3: LONG ANSWER QUESTION (20 MARKS)

49. a) Draw the output characteristic of a common emitter configuration and indicate the following:

i. Saturation region **(1 mark)**

ii. Cut-off region **(1 mark)**

iii. Active region **(1 mark)**

iv. Breakdown region **(1 mark)**

v. DC load line **(1 mark)**

b) Draw a circuit a well labelled diagram of an asatable multi-vibrator. **(5 marks)**

c) i. State any three types of oscillators. **(3 marks)**

ii. State any two applications of oscillators. **(2 marks)**

d) Draw the truth table of the following logic gates:

i. EXOR gate **(2 marks)**

ii. EXNOR gate **(2 marks)**

iii. NOT gate **(1 mark)**